

Q1 - 24 January - Shift 2

Let the six numbers $a_1, a_2, a_3, a_4, a_5, a_6$ be in A.P. and $a_1 + a_3 = 10$. If the mean of these six numbers is $\frac{19}{2}$ and their variance is σ^2 , then $8\sigma^2$ is equal to

- (1) 220
- (2) 210
- (3) 200
- (4) 105

Space for your notes:

Q2 - 25 January - Shift 1

The mean and variance of the marks obtained by the students in a test are 10 and 4 respectively. Later, the marks of one of the students is increased from 8 to 12. If the new mean of the marks is 10.2, then their new variance is equal to :

- (1) 4.04
- (2) 4.08
- (3) 3.96
- (4) 3.92

Space for your notes:

Q3 - 29 January - Shift 1

There rotten apples are mixed accidentally with seven good apples and four apples are drawn one by one without replacement. Let the random variable X denote the number of rotten apples. If μ and σ^2 represent mean and variance of X , respectively, then $10(\mu^2 + \sigma^2)$ is equal to

(1) 20

(2) 250

(3) 25

(4) 30

Space for your notes:

Q4 - 29 January - Shift 2

Let $X = \{11, 12, 13, \dots, 40, 41\}$ and $Y = \{61, 62, 63, \dots, 90, 91\}$ be the two sets of observations. If

$\bar{x}$ and $\bar{y}$ are their respective means and σ^2 is the variance of all the observations in $X \cup Y$, then $|\bar{x} + \bar{y} - \sigma^2|$ is equal to _____.

Space for your notes:

Q5 - 30 January - Shift 1

The mean and variance of 7 observations are 8 and 16 respectively. If one observation 14 is omitted and a and b are respectively mean and variance of remaining 6 observation, then $a + 3b - 5$ is equal to _____.

Space for your notes:

Q6 - 30 January - Shift 2

Let S be the set of all values of a_1 for which the mean deviation about the mean of 100 consecutive positive integers $a_1, a_2, a_3, \dots, a_{100}$ is 25. Then S is

- (1) ϕ (2) $\{99\}$
 (3) $\mathbb{N}$ (4) $\{9\}$

Space for your notes:

Q7 - 31 January - Shift 1

If the variance of the frequency distribution

x_i	2	3	4	5	6	7	8
Frequency f_i	3	6	16	α	9	5	6

Space for your notes:

Q8 - 31 January - Shift 2

Let the mean and standard deviation of marks of class A of 100 students be respectively 40 and $\alpha (> 0)$, and the mean and standard deviation of marks of class B of n students be respectively 55 and $30 - \alpha$. If the mean and variance of the marks of the combined class of $100 + n$ students are respectively 50 and 350, then the sum of variances of classes A and B is:

- (1) 500 (2) 650
 (3) 450 (4) 900

Space for your notes:

Q9 - 01 February - Shift 1

The mean and variance of 5 observations are 5 and 8 respectively. If 3 observations are 1, 3, 5, then the sum of cubes of the remaining two observations is

- (1) 1072 (2) 1792
 (3) 1216 (4) 1456

Space for your notes:

Q10 - 01 February - Shift 2

Let $9 = x_1 < x_2 < \dots < x_7$ be in an A.P. with common difference d . If the standard deviation of $x_1, x_2, \dots, x_7$ is 4 and the mean is $\bar{x}$, then $\bar{x} + x_6$ is equal to :

Space for your notes:

(1) $18\left(1 + \frac{1}{\sqrt{3}}\right)$

(2) 34

(3) $2\left(9 + \frac{8}{\sqrt{7}}\right)$

(4) 25

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (2)

Q2 (3)

Q3 (1)

Q4 (603)

Q5 (37)

Q6 (3)

Q7 (5)

Q8 (1)

Q9 (1)

Q10 (2)

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Q1 (2)

$$a_1 + a_3 = 10 = a_1 + d \Rightarrow 5$$

$$a_1 + a_2 + a_3 + a_4 + a_5 + a_6 = 57$$

$$\Rightarrow \frac{6}{2}[a_1 + a_6] = 57$$

$$\Rightarrow a_1 + a_6 = 19$$

$$\Rightarrow 2a_1 + 5d = 19 \text{ and } a_1 + d = 5$$

$$\Rightarrow a_1 = 2, d = 3$$

Numbers : 2, 5, 8, 11, 14, 17

Variance = σ^2 = mean of squares – square of mean

$$= \frac{2^2 + 5^2 + 8^2 + (11)^2 + (14)^2 + (17)^2}{6} - \left(\frac{19}{2}\right)^2$$

$$= \frac{699}{6} - \frac{361}{4} = \frac{105}{4}$$

$$8\sigma^2 = 210$$

Q2 (3)

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$$\sum_{i=1}^n x_i = 10n$$

$$\sum_{i=1}^n x_i - 8 + 12 = (10.2)n \quad \therefore n = 20$$

$$\text{Now } \frac{\sum_{i=1}^{20} x_i^2}{20} - (10)^2 = 4 \Rightarrow \sum_{i=1}^{20} x_i^2 = 2080$$

$$\frac{\sum_{i=1}^{20} x_i^2 - 8^2 + 12^2}{20} - (10.2)^2$$

$$= 108 - 104.04 = 3.96$$

Q3 (1)

X	P(x)	XP(X)	X ² P(X)
0	1/6	0	0
1	1/2	1/2	1/2
2	3/10	6/10	12/10
3	1/30	1/10	9/30

$$\sum xP(x) = \frac{6}{2} = \mu$$

$$\sigma^2 = \sum x^2 P(x) - \mu^2$$

$$\sigma^2 + \mu^2 = 0 + \frac{1}{2} + \frac{12}{10} + \frac{9}{30} = 2$$

$$10(\sigma^2 + \mu^2) = 20 \text{ Ans.}$$

Q4 (603)

$$\bar{x} = \frac{\sum_{i=1}^{31} i}{31} = \frac{11+41}{2} = 26 \quad (31 \text{ elements})$$

$$\bar{y} = \frac{\sum_{j=61}^{91} j}{31} = \frac{61+91}{2} = 76 \quad (31 \text{ elements})$$

$$\begin{aligned} \text{Combined mean, } \mu &= \frac{31 \times 26 + 31 \times 76}{31 + 31} \\ &= \frac{26 + 76}{2} = 51 \end{aligned}$$

$$\sigma^2 = \frac{1}{62} \times \left(\sum_{i=1}^{31} (x_i - \mu)^2 + \sum_{i=1}^{31} (y_i - \mu)^2 \right) = 705$$

Since, $x_i \in X$ are in A.P. with 31 elements & common difference 1, same is $y_i \in y$, when written in increasing order.

$$\begin{aligned} \therefore \sum_{i=1}^{31} (x_i - \mu)^2 &= \sum_{i=1}^{31} (y_i - \mu)^2 \\ &= 10^2 + 11^2 + \dots + 40^2 \end{aligned}$$

$$= \frac{40 \times 41 \times 81}{6} - \frac{9 \times 10 \times 19}{6} = 21855$$

$$\therefore |\bar{x} + \bar{y} - \sigma^2| = |26 + 76 - 705| = 603$$

Q5 (37)

$$\frac{x_1 + x_2 + \dots + x_7}{7} = 8$$

$$\frac{x_1 + x_2 + x_3 + \dots + x_6 + 14}{7} = 8$$

$$\Rightarrow x_1 + x_2 + \dots + x_6 = 42$$

$$\therefore \frac{x_1 + x_2 + \dots + x_6}{6} = \frac{42}{6} = 7 = a$$

$$\frac{\sum x_i^2}{7} - 8^2 = 16$$

$$\sum x_i^2 = 560$$

$$\Rightarrow x_1^2 + x_2^2 + \dots + x_6^2 = 364$$

$$b = \frac{x_1^2 + x_2^2 + \dots + x_6^2}{6} - 7^2$$

$$= \frac{364}{6} - 49$$

$$b = \frac{70}{6}$$

$$a + 3b - 5 = 7 + 3 \times \frac{70}{6} - 5$$

$$= 37$$

Q6 (3)

let a_1 be any natural number $a_1, a_1 + 1, a_1 + 2, \dots, a_1 + 99$ are values of a_i 's

$$\bar{x} = \frac{a_1 + (a_1 + 1) + (a_1 + 2) + \dots + a_1 + 99}{100}$$

$$= \frac{100a_1 + (1 + 2 + \dots + 99)}{100} = a_1 + \frac{99 \times 100}{2 \times 100}$$

$$= a_1 + \frac{99}{2}$$

$$\text{Mean deviation about mean} = \frac{\sum_{i=1}^{100} |x_i - \bar{x}|}{100}$$

$$= \frac{2 \left(\frac{99}{2} + \frac{97}{2} + \frac{95}{2} + \dots + \frac{1}{2} \right)}{100}$$

$$= \frac{1 + 3 + \dots + 99}{100}$$

$$= \frac{\frac{50}{2} [1 + 99]}{100}$$

$$= 25$$

Q7 (5)

x_i	f_i	$d_i = x_i - 5$	$f_i d_i^2$	$f_i d_i$
2	3	-3	27	-9
3	6	-2	24	-12
4	16	-1	16	-16
5	α	0	0	0
6	9	1	9	9
7	5	2	20	10
8	6	3	54	18

$$\sigma_x^2 = \sigma_d^2 = \frac{\sum f_i d_i^2}{\sum f_i} - \left(\frac{\sum f_i d_i}{\sum f_i} \right)^2$$

$$= \frac{150}{45 + \alpha} - 0 = 3$$

$$\Rightarrow 150 = 135 + 3\alpha$$

$$\Rightarrow 3\alpha = 15 \Rightarrow \alpha = 5$$

Q8 (1)

A	B	A+B
$\bar{x}_1 = 40$	$\bar{x}_2 = 55$	$\bar{x} = 50$
$\sigma_1^2 = \alpha$	$\sigma_2^2 = 30 - \alpha$	$\sigma^2 = 350$
$n_1 = 100$	$n_2 = n$	$100 + n$
$\bar{x} = \frac{100 \times 40 + 55n}{100 + n}$		
$5000 + 50n = 4000 + 55n$		
$1000 = 5n$		
$n = 200$		
$\sum x_i^2$		
$\sigma_1^2 = \frac{\sum x_i^2}{100} - 40^2$		
$\sum x_j^2$		
$\sigma_2^2 = \frac{\sum x_j^2}{100} - 55^2$		
$350 = \sigma^2 = \frac{\sum x_i^2 + \sum x_j^2}{300} - (\bar{x})^2$		
$350 = \frac{(1600 + \alpha^2) \times 100 + [(30 - \alpha)^2 + 3025] \times 200}{300} - (50)^2$		
$2850 \times 3 = \alpha^2 + 2(30 - \alpha)^2 + 1600 + 6050$		
$8550 = \alpha^2 + 2(30 - \alpha)^2 + 7650$		
$\alpha^2 + 2(30 - \alpha)^2 = 900$		
$\alpha^2 - 40\alpha + 300 = 0$		
$\alpha = 10, 30$		
$\sigma_1^2 + \sigma_2^2 = 10^2 + 20^2 = 500$		

Q9 (1)

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$$\frac{1+3+5+a+b}{5} = 5$$

$$a+b=16 \dots\dots(1)$$

$$\sigma^2 = \frac{\sum x_1^2}{5} - \left(\frac{\sum x}{5} \right)^2$$

$$8 = \frac{1^2+3^2+5^2+a^2+b^2}{5} - 25$$

$$a^2+b^2=130 \dots\dots(2)$$

by (1), (2)

$$a=7, b=9$$

Q10 (2)

$$9 = x_1 < x_2 < \dots < x_7$$

$$9, 9 + d, 9 + 2d, \dots, 9 + 6d$$

$$0, d, 2d, \dots, 6d$$

$$\bar{x}_{\text{new}} = \frac{21d}{7} = 3d$$

$$16 = \frac{1}{7}(0^2 + 1^2 + \dots + 6^2)d^2 - 9d^2$$

$$= \frac{1}{7} \left(\frac{6 \times 7 \times 13}{6} \right) d^2 - 9d^2$$

$$16 = 4d^2$$

$$d^2 = 4$$

$$d = 2$$

$$\bar{x} + x_6 = 6 + 9 + 10 + 9$$